Baloyi Thabang Bongani Junior

Data Science | *Machine Learning* | *Mathematical Statistics*

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PROFESSIONAL SUMMARY

Data scientist with an MSc in Mathematical Statistics completed with distinction and an Honours degree in Statistics and Data Science. More than five years of experience building, evaluating, and deploying machine learning models, from exploratory analysis and feature engineering through to production APIs, monitoring, and stakeholder reporting, across healthcare, fitness, e-commerce, and enterprise planning.

Strong in Python, SQL, R, TensorFlow, Keras, scikit-learn, Spark, and Databricks. Combines statistical depth in simulation, time series, and Bayesian analysis with the data engineering, RAG, and CI/CD practice needed to take models into production and keep them reliable.

EDUCATION

University of the Free State — **Jan 2026 to Oct 2026**
Master of Science, Mathematical Statistics — *80% (With Distinction)* | *Academic Record*

University of Cape Town — **Dec 2025**
BSc Honours, Statistics and Data Science, minor in Actuarial Science — *74%* | *Transcript* | *Proof of Qualification*

University of Cape Town — **Dec 2022**
Bachelor of Commerce, Management Studies — *70%* | *Transcript* | *Certificate*

University of the Free State — **2016 to 2017**
Bachelor of Science, Actuarial Sciences (Incomplete) — *Not completed* | *Academic Record*

RESEARCH

- **Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values** (MSc dissertation, UFS, 2026): built a three-stage framework on a one-million-record benchmark dataset. Selected a deep neural network through a five-phase tuning process and compared it with a traditional pricing model, Random Forest, and gradient boosting on held-out data. Used Monte Carlo simulation to generate a full predicted cost distribution with validation diagnostics, then applied threshold-conditional Shapley attribution to show that upper-tail cost drivers differ from average drivers.
- **Mean-Variance Hedging with Neural Networks using VAR Sample Paths** (BSc Honours thesis, UCT, 2024, co-authored): trained a feedforward network on VAR(11) simulated price paths to hedge a spread option, and benchmarked P&L against Black-Scholes delta hedging on 5,000 simulated paths and on market data: the network earned a higher mean P&L, while delta hedging delivered the lower P&L variance.

TECHNICAL SKILLS

Machine Learning: Deep Learning, Neural Networks, Random Forests, XGBoost, Logistic Regression, Classification, Regression, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Model Monitoring, Shapley Explainability

Statistics & Optimisation: Statistical Modelling, Hypothesis Testing, Time Series Analysis, Multivariate Analysis, Bayesian Analysis, Monte Carlo Simulation, Linear and Mixed-Integer Programming

Programming: Python, SQL, R, MATLAB, STATA, C#, Java, VBA, NumPy, pandas, scikit-learn, TensorFlow, Keras, matplotlib, pytest

LLM Applications: RAG Pipelines, Embeddings, Semantic Search, pgvector, Azure OpenAI, Prompt Engineering, Multi-Agent Systems, n8n

Data & BI: PostgreSQL, Snowflake, Amazon Redshift, Spark, PySpark, Databricks, Delta Lake, dbt, ETL Pipelines, Data Validation, Power BI, Tableau, Excel

Cloud & Delivery: Azure, AWS, Docker, Git, GitHub Actions, Azure DevOps, Bitbucket Pipelines, SonarQube, Flask, REST APIs

WORK EXPERIENCE

Senior Full Stack AI Engineer
4S4, Cape Town, ZA (Full Time, Hybrid) *Jun 2026 to Present*

- **Clients:** Rheinmetall Denel Munition, Old Mutual Bank, Ooba, Deloitte
- Implemented Tiffany, an embedded voice and text AI planning co-pilot integrating OpenAI speech and realtime services, registered planning commands, structured outputs, audit trails, and human confirmation for governed actions
- Build and orchestrate agentic and automation workflows using n8n, multi-agent systems, and RAG pipelines
- Led development of EAPP, an enterprise planning platform integrating SAP-derived demand, stock and open supply, multi-level BOM explosion, finite capacity, and cross-plant scheduling
- Built explainable decision workflows using Monte Carlo lead-time simulation, critical-path analysis, linear programming, and lexicographic mixed-integer programming

Artificial Intelligence Engineer
Full Stack, Cape Town, ZA (Full Time) *Mar 2023 to May 2026*

- **Clients:** Virgin Active UK, Virgin Active SA, Expandly
- Design and build AI-driven, cloud-native applications on AWS and Azure, integrating LLM services, data workflows, and Keras-based models into production systems
- Develop backend systems and data services for RAG pipelines, embeddings, semantic search, recommendation workflows, and analytics-ready outputs
- Build data and ML workflows using Python, SQL, Spark, PySpark, Databricks, PostgreSQL, pgvector, and Azure OpenAI
- Integrate ML models into Flask services and production APIs with monitoring, validation checks, and data quality controls
- Build AI safety components, including guardrails, risk-pattern detection, medical caution logic, and auditable decision workflows

AI Machine Learning Engineer Intern
Full Stack, Cape Town, ZA *Jan 2023 to Mar 2023*

- Performed exploratory data analysis on e-commerce and member engagement datasets using pandas, NumPy, matplotlib, and SQL
- Built and evaluated classification and regression models, including Logistic Regression, Random Forests, and XGBoost
- Developed preprocessing and feature engineering pipelines for missing values, categorical encoding, scaling, and data validation

Junior Data Scientist
Discovery Limited, Cape Town, ZA (Full Time) *Jan 2022 to Dec 2022*

- Developed, trained, and evaluated machine learning models using Python and SQL on large-scale healthcare and operational datasets to optimise client scheme services.
- Engineered features from structured and unstructured data sources, implementing predictive models to enhance member engagement and operational workflows.
- Established model monitoring protocols to track drift, performance degradation, and data pipeline integrity across production database environments.
- Collaborated with clinical, actuarial, and business stakeholders to translate domain requirements into statistical solutions and communicated quantitative insights to non-technical users.

Data Analyst Intern
Discovery Limited, Cape Town, ZA *Jan 2021 to Dec 2021*

- Analysed large customer datasets using Advanced Excel and SQL to extract behavioural patterns and deliver actionable insights for customer engagement strategies.
- Designed automated dashboards and recurring analytical reports to present core customer and health engagement metrics to stakeholders.

PROFESSIONAL QUALIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate (DP-700), 2026; CFA Level I (In Progress)
- Actuarial Society of South Africa exemptions: A112 (Business Economics), A113 (Business Finance)

HONOURS & AWARDS

- Best Aggregate Average Actuarial Science and Dean's List, UFS; Golden Key; Moshal and Jurgen Klauss Bathe Scholarships